RHCSA Study Guide

Red Hat Certified System Administrator — EX200 Exam

1. Understand and Use Essential Tools

Access a shell prompt and issue commands with correct syntax

Use the bash shell to run commands. Understand command structure: command [options] [arguments].

• Tab completion helps complete commands and file names

• Use man for manual pages

• Use --help flag for quick help

• history command shows previous commands

• !! reruns last command

Key Commands:

```
ls -la /etc
man passwd
history
echo $PATH
```

Use input-output redirection

> redirects stdout to a file (overwrites)

>> appends stdout to a file

< redirects file as stdin

2> redirects stderr

&> redirects both stdout and stderr

| pipes output of one command to another

Key Commands:

```
echo 'hello' > file.txt
ls /etc >> output.txt
grep 'error' /var/log/messages 2>/dev/null
cat file.txt | grep keyword
```

Use grep and regular expressions

grep searches for patterns in files.

-i = case insensitive, -r = recursive, -n = line numbers, -v = invert match, -E = extended regex

Key Commands:

```
grep 'root' /etc/passwd
grep -i 'error' /var/log/messages
```

```
grep -rn 'sshd' /etc/
grep -E '^[0-9]+' file.txt
```

Access remote systems using SSH

SSH provides secure remote login.

• ssh user@hostname — connect to remote host

• ssh-keygen — generate key pair

• ssh-copy-id — copy public key to remote host

• scp — secure copy files

Key Commands:

```
ssh admin@192.168.1.10
ssh-keygen -t rsa -b 4096
ssh-copy-id admin@192.168.1.10
scp file.txt admin@host:/tmp/
```

Log in and switch users

• su - username — switch to another user (full environment)

• sudo command — run command as root

• whoami — show current user

• id — show user and group info

Key Commands:

```
su - admin
sudo bash
whoami
id
```

Archive, compress, unpack files

tar is the main archiving tool.

• -c = create, -x = extract, -v = verbose, -f = filename

• -z = gzip compression, -j = bzip2, -J = xz

Key Commands:

```
tar -czvf archive.tar.gz /etc/
tar -xzvf archive.tar.gz
tar -cjvf archive.tar.bz2 /home/
zip -r files.zip /var/www/
```

Create and edit text files

Key editors: vim and nano

Vim basics: i=insert, Esc=normal mode, :w=save, :q=quit, :wq=save+quit, :q!=force quit

Nano: Ctrl+O=save, Ctrl+X=exit

Key Commands:

```
vim /etc/hosts
nano /etc/hostname
cat > newfile.txt
touch emptyfile.txt
```

Create, delete, copy, move files and directories

• mkdir — make directory

• rm — remove files (-r for recursive, -f for force)

• cp — copy (-r for directories)

• mv — move or rename

• find — search for files

Key Commands:

```
mkdir -p /data/project/logs
rm -rf /tmp/olddir
cp -r /etc/skel /home/newuser
mv oldname.txt newname.txt
find / -name '*.conf' -type f
```

2. Create Simple Shell Scripts

Write simple shell scripts

Scripts start with a shebang line. Make executable with chmod +x.

Key concepts: variables, conditionals, loops, functions.

Key Commands:

```
#!/bin/bash
# Simple backup script
BACKUP_DIR=/backup
DATE=$(date +%Y%m%d)
mkdir -p $BACKUP_DIR
tar -czf $BACKUP_DIR/home_$DATE.tar.gz /home/
echo 'Backup complete'
```

Conditionals and loops

if/then/else for decisions. for and while for loops.

Key Commands:

```
# If statement
if [ -f /etc/passwd ]; then
echo 'File exists'
fi

# For loop
for user in alice bob charlie; do
useradd $user
done

# While loop
```

```
while read line; do
echo $line
done < file.txt
```

3. Operate Running Systems

Boot, reboot, and shut down a system

• systemctl reboot — restart the system

• systemctl poweroff — shut down

• systemctl halt — halt without powering off

• shutdown -h now — immediate shutdown

• shutdown -r +5 — reboot in 5 minutes

Key Commands:

```
systemctl reboot
systemctl poweroff
shutdown -h now
shutdown -r +10 'Rebooting in 10 min'
```

Boot systems into different targets

Targets replace runlevels in systemd.

• multi-user.target = runlevel 3 (no GUI)

• graphical.target = runlevel 5 (GUI)

• rescue.target = single user mode

• emergency.target = minimal environment

Key Commands:

```
systemctl get-default
systemctl set-default multi-user.target
systemctl isolate rescue.target
# At boot: append 'systemd.unit=rescue.target' to kernel line
```

Interrupt boot to gain root access

At GRUB menu: press 'e' to edit, find linux line, append 'rd.break' or 'init=/bin/bash', Ctrl+X to boot. Then remount and chroot to reset root password.

Key Commands:

```
# After booting with rd.break:
mount -o remount,rw /sysroot
chroot /sysroot
passwd root
touch /.autorelabel
exit
reboot
```

Identify CPU and memory usage

• top / htop — real-time process viewer

• ps aux — snapshot of processes

• free -h — memory usage

• uptime — load averages

• vmstat — virtual memory stats

Key Commands:

```
top
ps aux | grep httpd
free -h
uptime
kill -9
killall httpd
```

Adjust process scheduling

nice sets priority at launch (-20 highest, 19 lowest).

renice changes priority of running process.

Key Commands:

```
nice -n 10 tar -czf archive.tar.gz /home
renice -n 5 -p 1234
ps -eo pid,ni,comm | grep tar
```

Manage tuning profiles

tuned optimizes system performance for different workloads.

Key Commands:

```
tuned-adm list
tuned-adm active
tuned-adm profile throughput-performance
tuned-adm recommend
```

4. Configure Local Storage

List, create, delete partitions

• fdisk — MBR partition tool

• gdisk — GPT partition tool

• parted — supports both

• lsblk — list block devices

• blkid — show UUIDs

Key Commands:

```
lsblk
fdisk /dev/sdb
gdisk /dev/sdb
parted /dev/sdc print
blkid /dev/sdb1
```

Create and remove physical volumes, volume groups, logical volumes

LVM provides flexible storage management.

• PV = Physical Volume (raw disk/partition)

• VG = Volume Group (pool of PVs)

• LV = Logical Volume (usable storage)

Key Commands:

```
pvcreate /dev/sdb1
vgcreate datavg /dev/sdb1
lvcreate -L 10G -n datalv datavg
lvextend -L +5G /dev/datavg/datalv
resize2fs /dev/datavg/datalv
pvs && vgs && lvs
```

Configure systems to mount filesystems at boot

/etc/fstab controls what mounts at boot.

Format: device mountpoint fstype options dump pass

Key Commands:

```
# /etc/fstab entry:
/dev/sdb1 /data xfs defaults 0 0
UUID=abc123 /mnt/data ext4 defaults 0 2

mount -a # test fstab entries
mount /dev/sdb1 /data
```

Create and use swap space

Swap extends RAM using disk space.

Key Commands:

```
mkswap /dev/sdb2
swapon /dev/sdb2
free -h
# Add to /etc/fstab:
/dev/sdb2 swap swap defaults 0 0

# Create swap file:
dd if=/dev/zero of=/swapfile bs=1M count=1024
chmod 600 /swapfile
mkswap /swapfile
swapon /swapfile
```

5. Create and Configure File Systems

Create, mount, unmount file systems

Common filesystems: xfs (default RHEL), ext4, vfat

• mkfs.xfs — format as XFS

• mkfs.ext4 — format as ext4

• mount / umount — mount and unmount

Key Commands:

```
mkfs.xfs /dev/sdb1
mkfs.ext4 /dev/sdc1
mount /dev/sdb1 /mnt/data
umount /mnt/data
df -hT
```

Mount NFS network shares

NFS allows mounting remote filesystems.

Key Commands:

```
showmount -e 192.168.1.10
mount -t nfs 192.168.1.10:/exports/data /mnt/nfs
# Persistent in /etc/fstab:
192.168.1.10:/exports/data /mnt/nfs nfs defaults 0 0
```

Extend existing logical volumes

LVM allows online extension of volumes.

Key Commands:

```
lvextend -L +10G /dev/datavg/datalv
# For XFS:
xfs_growfs /dev/datavg/datalv
# For ext4:
resize2fs /dev/datavg/datalv
```

Configure autofs

autofs mounts filesystems on demand.

Key Commands:

```
dnf install autofs
systemctl enable --now autofs
# /etc/auto.master:
/misc /etc/auto.misc
# /etc/auto.misc:
cd -fstype=iso9660,ro,nosuid,nodev :/dev/cdrom
nfs 192.168.1.10:/exports/data
```

Set and configure disk quotas

Quotas limit disk usage per user/group.

Key Commands:

```
# Add usrquota to /etc/fstab options
mount -o remount /home
```

```
quotacheck -cug /home
quotaon /home
edquota -u username
repquota /home
```

Manage layered storage (Stratis)

Stratis is a modern storage management solution in RHEL 9.

Key Commands:

```
dnf install stratisd stratis-cli
systemctl enable --now stratisd
stratis pool create mypool /dev/sdb
stratis filesystem create mypool myfs
mount /stratis/mypool/myfs /mnt/stratis
```

6. Deploy, Configure, and Maintain Systems

Schedule tasks using cron and at

cron — recurring scheduled tasks

at — one-time scheduled tasks

crontab format: min hour day month weekday command

Key Commands:

```
crontab -e
# Run backup at 2am daily:
0 2 * * * /scripts/backup.sh
# Run every Monday at 9am:
0 9 * * 1 /scripts/report.sh

at 2pm tomorrow
at> /scripts/task.sh
at> Ctrl+D
atq # list pending jobs
```

Start and stop services, configure services at boot

systemctl manages services in RHEL.

Key Commands:

```
systemctl start httpd
systemctl stop httpd
systemctl restart httpd
systemctl reload httpd
systemctl enable httpd
systemctl disable httpd
systemctl status httpd
systemctl is-active httpd
systemctl list-units --type=service
```

Configure time service clients

chrony is the default NTP client in RHEL.

Key Commands:

```
timedatectl status
timedatectl set-timezone Asia/Tokyo
timedatectl set-ntp true
chronyc sources
chronyc tracking
# Edit /etc/chrony.conf to add NTP servers:
server time.cloudflare.com iburst
```

Install and update software packages

dnf is the package manager in RHEL 8/9.

Key Commands:

```
dnf install httpd
dnf remove httpd
dnf update
dnf update httpd
dnf search nginx
dnf info vim
dnf list installed
dnf provides /usr/bin/vim
dnf history
```

Manage repositories

Repos provide packages for dnf to install.

Key Commands:

```
dnf repolist
dnf repolist all
dnf config-manager --enable repo-name
dnf config-manager --add-repo URL
# Create /etc/yum.repos.d/custom.repo:
[custom]
name=Custom Repo
baseurl=http://repo.example.com/
enabled=1
gpgcheck=0
```

Work with package module streams

AppStream modules allow multiple versions of software.

Key Commands:

```
dnf module list
dnf module list nodejs
dnf module enable nodejs:16
dnf module install nodejs:16
dnf module disable nodejs:16
dnf module reset nodejs
```

7. Manage Basic Networking

Configure IPv4 and IPv6 addresses

nmcli is the primary tool for network management in RHEL.

Key Commands:

```
nmcli device status
nmcli connection show
nmcli connection add type ethernet con-name eth0 ifname eth0
nmcli connection modify eth0 ipv4.addresses 192.168.1.100/24
nmcli connection modify eth0 ipv4.gateway 192.168.1.1
nmcli connection modify eth0 ipv4.dns 8.8.8.8
nmcli connection modify eth0 ipv4.method manual
nmcli connection up eth0
```

Configure hostname and name resolution

• hostnamectl sets the system hostname

• /etc/hosts for local name resolution

• /etc/resolv.conf for DNS servers

Key Commands:

```
hostnamectl set-hostname server1.example.com
hostnamectl status
# Edit /etc/hosts:
192.168.1.10 server1.example.com server1
# Edit /etc/resolv.conf:
nameserver 8.8.8.8
nameserver 8.8.4.4
```

Configure network services to start automatically

Enable NetworkManager and services to auto-start.

Key Commands:

```
systemctl enable --now NetworkManager
nmcli connection modify eth0 connection.autoconnect yes
nmcli connection up eth0
```

Restrict network access using firewall

firewalld manages the firewall in RHEL.

Key Commands:

```
systemctl enable --now firewalld
firewall-cmd --state
firewall-cmd --list-all
firewall-cmd --add-service=http --permanent
firewall-cmd --add-service=https --permanent
firewall-cmd --add-port=8080/tcp --permanent
firewall-cmd --remove-service=ftp --permanent
firewall-cmd --reload
```

8. Manage Users and Groups

Create, delete, and modify local user accounts

• useradd — create user

• usermod — modify user

• userdel — delete user

• passwd — set password

• /etc/passwd and /etc/shadow store user info

Key Commands:

```
useradd -m -s /bin/bash alice
useradd -u 1500 -g developers bob
passwd alice
usermod -aG wheel alice
usermod -s /sbin/nologin serviceuser
userdel -r alice # -r removes home dir
chage -l alice # show password aging
```

Change passwords and password aging

chage controls password expiration policies.

Key Commands:

```
passwd username
chage -M 90 alice # max 90 days
chage -m 7 alice # min 7 days
chage -W 14 alice # warn 14 days before
chage -E 2024-12-31 alice # expire date
chage -d 0 alice # force change at next login
```

Create, delete, and modify local groups

• groupadd — create group

• groupmod — modify group

• groupdel — delete group

• /etc/group stores group info

Key Commands:

```
groupadd developers
groupadd -g 2000 admins
groupmod -n newname oldname
groupdel developers
usermod -aG developers alice
id alice
groups alice
```

Configure superuser access

The wheel group grants sudo access in RHEL.

Key Commands:

```
usermod -aG wheel username
# Or edit /etc/sudoers via visudo:
username ALL=(ALL) ALL
# Passwordless sudo:
username ALL=(ALL) NOPASSWD: ALL
# Command-specific sudo:
username ALL=(ALL) /sbin/shutdown,/sbin/reboot
```

9. Manage Security

Configure firewall settings using firewall-cmd

firewalld uses zones to group rules. Default zone is 'public'.

Key Commands:

```
firewall-cmd --get-zones
firewall-cmd --get-default-zone
firewall-cmd --set-default-zone=trusted
firewall-cmd --zone=public --add-service=http --permanent
firewall-cmd --zone=public --add-port=443/tcp --permanent
firewall-cmd --zone=public --add-source=192.168.1.0/24 --permanent
firewall-cmd --reload
firewall-cmd --list-all
```

Manage default file permissions

umask controls default permissions for new files.

• Default file: 666 - umask

• Default dir: 777 - umask

• Common umask: 022 (files=644, dirs=755)

Key Commands:

```
umask
umask 022
umask 027
# Set permanently in ~/.bashrc or /etc/profile
echo 'umask 022' >> /etc/profile
stat filename # show permissions
```

Configure key-based authentication for SSH

SSH keys are more secure than passwords.

Key Commands:

```
ssh-keygen -t rsa -b 4096 -C 'user@host'
ssh-copy-id user@remotehost
# Manual method:
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
# Disable password auth in /etc/ssh/sshd_config:
PasswordAuthentication no
```

Set enforcing and permissive modes for SELinux

SELinux provides mandatory access control. NEVER disable it on the exam — set to permissive instead.

Key Commands:

```
getenforce
setenforce 0 # permissive (temporary)
```

```
setenforce 1 # enforcing (temporary)
# Permanent change in /etc/selinux/config:
SELINUX=enforcing
# or
SELINUX=permissive
```

List and identify SELinux file and process context

SELinux contexts control access for files and processes.

Key Commands:

```
ls -Z /etc/passwd
ps -eZ | grep httpd
stat -c %C /var/www/html
id -Z
```

Restore default file contexts

restorecon resets SELinux contexts to defaults.

Key Commands:

```
restorecon -v /var/www/html/index.html
restorecon -Rv /var/www/html/
# Check what context should be:
semanage fcontext -l | grep '/var/www'
# Set custom context:
semanage fcontext -a -t httpd_sys_content_t '/mywebdir(/.*)?'
restorecon -Rv /mywebdir
```

Manage SELinux port labels

If running a service on a non-standard port, add SELinux port label.

Key Commands:

```
semanage port -l | grep http
semanage port -a -t http_port_t -p tcp 8888
semanage port -d -t http_port_t -p tcp 8888
# Allow httpd to connect to network:
setsebool -P httpd_can_network_connect on
getsebool -a | grep httpd
```

Use boolean settings to modify SELinux

SELinux booleans toggle specific permissions on/off.

Key Commands:

```
getsebool -a
getsebool httpd_can_network_connect
setsebool httpd_can_network_connect on
setsebool -P httpd_can_network_connect on # -P makes it permanent
semanage boolean -l | grep httpd
```

Diagnose and address SELinux policy violations

audit2why and audit2allow help diagnose and fix SELinux issues.

Key Commands:

```
ausearch -m avc -ts recent
sealert -a /var/log/audit/audit.log
audit2why < /var/log/audit/audit.log
audit2allow -a
# Check SELinux denials:
cat /var/log/audit/audit.log | grep denied
```

10. Manage Containers

Find and retrieve container images

podman is the default container tool in RHEL (no daemon required).

Key Commands:

```
podman search httpd
podman pull registry.access.redhat.com/ubi9/httpd-24
podman images
podman image inspect ubi9/httpd-24
```

Inspect container images

Inspect gives detailed info about images and containers.

Key Commands:

```
podman inspect image-name
podman history image-name
podman image ls
skopeo inspect docker://registry.access.redhat.com/ubi9
```

Perform container management

Basic container lifecycle commands.

Key Commands:

```
podman run -d --name myapp -p 8080:80 httpd
podman ps
podman ps -a
podman stop myapp
podman start myapp
podman restart myapp
podman rm myapp
podman exec -it myapp bash
podman logs myapp
```

Run a service inside a container

Run containers with persistent storage and network access.

Key Commands:

```
podman run -d --name webserver \
-p 8080:80 \
-v /host/html:/var/www/html:Z \
httpd
# Test:
```

```
curl http://localhost:8080
# :Z label fixes SELinux context for mounted volumes
```

Configure a container to start automatically as a systemd service

Generate systemd unit files from running containers.

Key Commands:

```
podman generate systemd --name myapp --files --new
mv container-myapp.service ~/.config/systemd/user/
systemctl --user daemon-reload
systemctl --user enable --now container-myapp
systemctl --user status container-myapp
# For system-wide (root):
cp container-myapp.service /etc/systemd/system/
systemctl daemon-reload
systemctl enable --now container-myapp
```

Attach persistent storage to a container

Volumes provide persistent data that survives container restarts.

Key Commands:

```
podman volume create mydata
podman run -d --name myapp \
-v mydata:/var/lib/myapp:Z \
myimage
podman volume ls
podman volume inspect mydata
podman volume rm mydata
```

Exam Tips & Tricks

■ Always use 'permanent' flag with firewall-cmd and reload after: --permanent then firewall-cmd --reload

■ Never disable SELinux — set to permissive if needed, never 'disabled'

■ Always enable services with --now to start immediately: systemctl enable --now httpd

■ Test /etc/fstab with 'mount -a' before rebooting to avoid boot failures

■ Use 'man' pages on the exam — they're available and contain examples

■ Verify your work after every task — check status, test connections, etc.

■ The exam is hands-on — practice these commands until they're muscle memory

■ Time management is key — don't spend too long on one task

■ Use tab completion to avoid typos in commands and paths

■ Remember: systemctl changes need --permanent equivalent or they reset on reboot